



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Computer Science & Engineering Department

Subject Name	Operating System	A.Y	2025-26
Subject Code	303105251	Semester	4th sem
Unit 1			
Sr No	Question	COs	B.T
1	Define an Operating System.	1	2
2	Explain the structure of layered operating system with diagram.	1	4
3	Explain various types of operating systems (Batch, Time Sharing, Multiprogramming, RTOS, Distributed OS) with examples.	1	2
4	Explain kernel mode and user mode of operation.	1	2
5	Discuss the role of an OS as a resource manager.	1	2
6	What is the difference between user mode and kernel mode?	1	2
7	Explain the concept of Virtual Machines. Compare Process VM and System VM.	1	3
8	Explain how interrupts are handled by an operating system.	1	2
9	What is a virtual machine? Explain its advantages.	1	2
10	Explain the differences between Batch OS and Time-Sharing OS.	1	3
11	Compare multiprogramming and multitasking with examples.	1	4
12	Explain system programs and their types.	1	2
13	Explain how virtualization improves hardware utilization.	1	3
14	Describe the main functions of the shell in an operating system.	1	2
15	Explain in detail the role of the kernel. Discuss its types and responsibilities.	1	4